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COMETFWC067

Multiplexer Boolean Expression – GATE 2010 CS Q9
Hardware Verification on Pygmy FPGA

Aim

To derive the simplified Boolean expression for the output f of the given 4:1 multiplexer (data inputs R, R', R', R; select lines P, Q) and to verify the derived logic on hardware by blinking an LED on the Vamon FPGA board according to the computed output.

GATE Question

The Boolean expression for the output f of a 4:1 multiplexer with inputs I0 = R, I1 = R', I2 = R', I3 = R and select lines S1 = P, S0 = Q is one of the following:

Option	Expression
A	(P xor Q xor R)'
B	P xor Q xor R
C	P + Q + R
D	(P + Q + R)'

Question Analysis

1. Select-Line to Output Mapping

S1 (P)	S0 (Q)	Line Selected	Output f
0	0	I0	R
0	1	I1	R'
1	0	I2	R'
1	1	I3	R

2. Sum of Products Form

```
f = P'Q'R + P'Q R' + PQ' R' + PQR
```

3. Complete Truth Table

P	Q	R	f

0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

4. Observation and Final Expression

f is 1 only where P, Q, R contain an odd number of 1's (001, 010, 100, 111) – the defining property of the three-input XOR operation.

```
f = P xor Q xor R → Answer: Option (B)
```

1. Components Used

Component
LED
Vamon Board
Jumper Wires
Linux Computer

2. Purpose of Each Component

Component	Function
Vamon Board	Implements the multiplexer logic (P xor Q xor R) and drives the LED
Jumper Wires	Connects the Vamon board output pin to the LED circuit
Linux Computer	Runs Termux/toolchain to compile and program the board
LED	Displays the logic output f for a given P, Q, R combination

3. Power Connections

Pin	Connection
Vamon Board Vcc	USB +5V (via onboard regulator)
Vamon Board GND	USB GND (common ground)

4. Vamon Board to LED Connections

Connection	Description
Vamon Board Pin (LED Output) -> Resistor	Current limiting resistor (220 ohm - 330 ohm)
Resistor -> LED Anode (+)	Connect resistor output to LED anode
LED Cathode (-) -> GND	Connect LED cathode to Ground
Vamon Board GND -> Circuit GND	Common ground reference

5. Execution Procedure

1. Write the multiplexer logic (f = P xor Q xor R) as a Verilog module directly inside Termux on the Linux computer, along with a pin-mapping constraint file for the board.

2. Run the SymbiFlow toolchain from Termux to synthesize the module and generate a bitstream for the target board.

3. Locate the generated bitstream and rename it to a .bin file so the board's programmer utility can read it.

4. Plug the Vamon board into the Linux computer through USB and switch it into flashing/programming mode.

5. Confirm the computer has detected the board as a serial device before proceeding.

6. Flash the .bin file onto the board using the TinyFPGA programmer utility, run from the computer.

7. Once flashing finishes, reset the board so the new logic takes effect.

8. Apply each combination of P, Q, and R and watch the LED: it should light exactly when f = P xor Q xor R evaluates to 1, confirming the design works as derived.

Results and Observations

The generated Verilog design was successfully created in Termux, compiled, and programmed onto the Vamon board through the connected Linux computer. After programming and resetting the board, the LED output responded according to the logic combinations applied at the inputs P, Q, and R, matching the truth table derived above for f = P xor Q xor R.

P	Q	R	LED Output (f)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0

1	1	0	0
1	1	1	1